

One-bit Phase Retrieval via Double-Sparsity Constraints

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Abstract—One-bit phase retrieval (1bPR) has been explored under various models. Conventional convex-lifted formulations do not incorporate sparsity and thus cannot be applied to sparse signals. Although one-sided ℓ_1 -loss formulation has been designed to handle sparse signals, its performance degrades when sign flips occur. In this work, we exploit the double-sparsity constrained optimization framework to improve reconstruction performance in the presence of noise. Then, we adopt projected gradient descent with backtracking and subspace pursuit to tackle the resultant problem. Compared with the ℓ_1 -loss based method, our algorithm has stronger robustness to additive noise. Experimental results verify its superior performance over the existing method.

Index Terms—One-bit sparse phase retrieval, robustness, projected gradient descent, subspace pursuit, random dithers.

I. INTRODUCTION

PHASE retrieval (PR) has broad applications in optics, X-ray crystallography, diffraction imaging, and wireless communications [1]–[9]. It is a nonlinear problem as the absolute value function is exploited to measure magnitude. In addition, the absence of phase information makes the feasible set nonconvex as multiple distinct signals can produce the same magnitude measurements. These issues lead to theoretical and computational challenges [10]. Several approaches were proposed to handle PR. The early contributions include iterative error reduction algorithms [7], [8]. In addition, [3] formulated PR as a convex semidefinite programming (SDP), while [11] provided a convex relaxation based on SDP. Moreover, scalable gradient-descent approaches, such as Wirtinger flow [2] and its variants [12]–[16], were proposed.

Recently, the compressed sensing (CS) technique has been applied for PR. Moravec et al. proposed compressive PR, which introduces a compressibility constraint on signals [17]. Shechtman et al. devised a greedy local search-based algorithm for sparse PR [18], which iteratively refines the signal support using Gauss-Newton update. Wang et al. combined truncated gradient iteration with adaptive support estimation to recover the phase of sparse signals [19]. Moreover, convex approaches via the lifting technique were extended to sparse signals, including sparse convex relaxation [20] and atomic matrix

norm regularization [21]. As mentioned in [22], the structural assumptions on the signal and the observations can speed up convergence and improve recovery performance.

Quantization is a necessary procedure in digital acquisition systems. One-bit quantization is advantageous due to its compatibility with ultra-low-cost and low-power analog-to-digital converters (ADCs) [23], [24]. However, retaining only the sign of measurements leads to severe magnitude loss [25].

Mukherjee and Seelamantula [26] formulated the task using the matrix format, while incorporates a one-sided quadratic consistency penalty. Then, they solved the resultant rank-constrained problem using accelerated projected gradient descent. In addition, Eamatz et al. [27] proposed the one-bit phase retrieval algorithm (OPeRA) that enforces one-bit consistency through projection onto the binary inequality constraint and benefits from a large number of sign measurements. With sufficient measurements, the resultant linear inequality forces the feasible set to lie inside the positive semidefinite (PSD) cone, making the rank and PSD constraints unnecessary. To process sparse signals, Chen and Yuan [28] interpreted one-bit measurements as a phaseless version of random hyperplane tessellation and proposed a one-sided ℓ_1 -loss minimization method for sparse signals. However, this method is sensitive to additive noise.

In this work, we formulate one-bit sparse phase retrieval (1bSPR) as a double-sparsity constrained optimization problem. In contrast to the one-sided ℓ_1 -loss programme, our formulation explicitly models sign-inconsistency via a sparse auxiliary variable to increase the recovery performance in the presence of noise. To solve the resultant nonconvex optimization problem, we combine projected gradient descent and subspace pursuit technique to propose an algorithm. Within the subspace pursuit step, a first-order linearization is applied to approximate the absolute-value term. Consequently, the linearization yields a locally quadratic approximation, such that the nonlinear restricted subproblem is converted into a tractable least-squares problem with a closed-form solution. Finally, numerical results verify that our approach achieves superior performance over the existing methods.

II. RELATED WORKS

The observation model of the one-bit phase retrieval (1bPR) is given by

$$c_i = \text{sgn}(|\mathbf{a}_i^\top \mathbf{x}| - \tau_i), \quad i = 1, 2, \dots, m, \quad (1)$$

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where $\mathbf{a}_i \in \mathbb{R}^n$ is a sensing vector, $\mathbf{x} \in \mathbb{R}^n$ is the target signal, τ_i is a known threshold, $\text{sgn}(\cdot)$ is the sign function, and $|\cdot|$ is the absolute value operator. Then, the matrix form is expressed as

$$\mathbf{c} = \text{sgn}(|\mathbf{A}\mathbf{x}| - \boldsymbol{\tau}), \quad (2)$$

where $\mathbf{A} = [\mathbf{a}_1, \dots, \mathbf{a}_m]^\top \in \mathbb{R}^{m \times n}$, $\boldsymbol{\tau} \in \mathbb{R}^m$, and $|\mathbf{A}\mathbf{x}| = [|\mathbf{a}_1^\top \mathbf{x}|, |\mathbf{a}_2^\top \mathbf{x}|, \dots, |\mathbf{a}_m^\top \mathbf{x}|]^\top$.

In [27], the one-bit constraint is formulated as

$$c_i(|\mathbf{a}_i^\top \mathbf{x}| - \tau_i) \geq 0, \quad i = 1, \dots, m. \quad (3)$$

Each inequality ensures that the reconstructed signal is consistent with the corresponding one-bit measurement. In practice, severe noise causes sign flips, which we term sign-inconsistency.

In [27], the signal variable is converted to a rank-one matrix, such that $\mathbf{X} = \mathbf{x}\mathbf{x}^\top$ and $\mathbf{V}_i = \mathbf{a}_i\mathbf{a}_i^\top$. Then, 1bPR is formulated as

$$\begin{aligned} \min_{\mathbf{X}} \quad & \text{rank}(\mathbf{X}) \\ \text{s.t.} \quad & c_i(\text{Tr}(\mathbf{V}_i\mathbf{X}) - \tau_i^2) \geq 0, \quad \forall i, \quad \mathbf{X} \succeq 0. \end{aligned} \quad (4)$$

Although this conversion renders the constraints affine, (4) is NP-hard caused by rank function. A common convex relaxation replaces rank function with nuclear norm. Then, (4) is converted to an equivalent PSD problem. However, solving the resultant SDP is computationally expensive for large-scale problems.

In the lifting framework, the binary phase retrieval (BPR) method [26] employs a one-sided quadratic penalty, leading to

$$\min_{\mathbf{X} \succeq 0} F(\mathbf{X}) \quad \text{s.t.} \quad \text{rank}(\mathbf{X}) = 1, \quad (5)$$

where $F(\mathbf{X}) = \sum_{i=1}^m f(c_i(\text{Tr}(\mathbf{V}_i\mathbf{X}) - \tau_i^2))$, $f(u) = \frac{1}{2}u^2\mathbf{1}_{(u \leq 0)}$. However, this formulation has a rank-one matrix constraint, which needs eigenvalue decomposition and leads to high computational complexity.

Another approach leverages the geometric properties of the feasible set under abundant measurements [27]. With sufficient measurements, the linear inequalities define a polyhedron

$$\mathcal{P} = \{\mathbf{X} \mid c_i(\text{Tr}(\mathbf{V}_i\mathbf{X}) - \tau_i^2) \geq 0\}, \quad (6)$$

which forms a finite-volume space that naturally shrinks into the PSD cone. Consequently, explicit rank and PSD constraints become redundant for recovering the unique solution. Exploiting this property, OPeRA adopts a randomized Kaczmarz update rule to iteratively project onto these linear constraints. However, in the undersampled regime, the constraints fail to form a bounded polyhedron, leading to reconstruction performance degradation.

Another representative example is the one-sided ℓ_1 -loss minimization method [28], and formulates the task as

$$\min_{\mathbf{x}} \quad \frac{1}{m} \sum_{i=1}^m \max\{0, -c_i(|\mathbf{a}_i^\top \mathbf{x}| - \tau_i)\}. \quad (7)$$

It admits a geometric interpretation via phaseless hyperplane separation and offers theoretical recovery guarantee under Gaussian measurements. However, its performance degrades in the presence of additive noise.

III. PROBLEM FORMULATION

In this work, we extend a double-sparsity constrained formulation [29] to the phaseless one-bit setting, leading to

$$\begin{aligned} \min_{\mathbf{x}, \mathbf{y}} \quad & \|\mathbf{c} \odot (|\mathbf{A}\mathbf{x}| - \boldsymbol{\tau}) + \mathbf{y} - \varepsilon_0 \mathbf{1}\|_2^2 + \eta \|\mathbf{x}\|_2^2, \\ \text{s.t.} \quad & \|\mathbf{x}\|_0 \leq s, \quad \|\mathbf{y}_+\|_0 \leq k, \end{aligned} \quad (8)$$

where $\mathbf{y} \in \mathbb{R}^m$ is an auxiliary variable to model sign-inconsistency, s specifies the upper bound sparsity level of $\mathbf{x}$, k is the upper bound sparsity of the positive part of $\mathbf{y}$, $\eta > 0$ is a regularization parameter, $\odot$ denotes the element-wise product, $\mathbf{1}$ is the all-ones vector, and ε_0 is a tolerance margin. The double-sparsity indicates that the sparsity constraint acts on $\mathbf{x}$ and $\mathbf{y}$ simultaneously. This is because $\mathbf{x}$ and sign flips in one-bit measurements are sparse.

To solve (8), we develop an algorithm using projected gradient descent and subspace pursuit [29]–[31].

A. Projected Gradient Descent

This subsection derives the projected gradient procedure. At the t -th iteration, we first compute the sub-gradient and the gradient as

$$\mathbf{u}^{(t)} = 2 \left[\mathbf{A}^\top ((\mathbf{c} \odot \text{sgn}(\mathbf{A}\mathbf{x}^{(t)})) \odot \mathbf{r}^{(t)}) + \eta \mathbf{x}^{(t)} \right], \quad (9a)$$

$$\mathbf{v}^{(t)} = 2 \cdot \mathbf{r}(\mathbf{x}^{(t)}, \mathbf{y}^{(t)}), \quad (9b)$$

Then, we employ the projected gradient descent with backtracking scheme to update with step-size $\alpha > 0$, given by

$$\mathbf{x}^{(t+1)} = \text{ProS}(\mathbf{x}^{(t)} - \alpha \mathbf{u}^{(t)}, s), \quad (10a)$$

$$\mathbf{y}^{(t+1)} = \text{ProK}(\mathbf{y}^{(t)} - \alpha \mathbf{v}^{(t)}, k), \quad (10b)$$

where ProS retains the s largest-magnitude entries of $\mathbf{x}$. In addition, ProK preserves all negative entries in $\mathbf{y}$. For the positive entries, if their total number is less than or equal to k , all of them are retained. Otherwise, only the k largest positive entries are kept, and the rest are set to zero.

B. Subspace Pursuit

This subsection performs a subspace pursuit procedure to increase computational efficiency. The main idea is to iteratively update the subsets of $\mathbf{x}$ and $\mathbf{y}$ if their support sets are fixed.

At the $(t+1)$ -th iteration, the optimization problem is reformulated as the following two subproblems

$$\begin{aligned} \hat{\mathbf{x}}_{\mathcal{T}}^{(t+1)} = \arg \min_{\mathbf{z} \in \mathbb{R}^{|\mathcal{T}|}} \quad & \left\{ \left\| \mathbf{c}_{\mathcal{T}} \odot (|\mathbf{A}_{\mathcal{T}, \mathcal{T}} \mathbf{z}| - \boldsymbol{\tau}_{\mathcal{T}}) \right. \right. \\ & \left. \left. + \mathbf{y}_{\mathcal{T}}^{(t+1)} - \varepsilon_0 \mathbf{1} \right\|_2^2 + \eta \|\mathbf{z}\|_2^2 \right\}, \end{aligned} \quad (11a)$$

$$\begin{aligned} \hat{\mathbf{y}}_{\mathcal{K}}^{(t+1)} = \arg \min_{\mathbf{w} \in \mathbb{R}^{|\mathcal{K}|}} \quad & \left\{ \left\| \mathbf{c}_{\mathcal{K}} \odot (|\mathbf{A}_{\mathcal{K}, \mathcal{T}} \hat{\mathbf{x}}_{\mathcal{T}}^{(t+1)}| - \boldsymbol{\tau}_{\mathcal{K}}) \right. \right. \\ & \left. \left. + \mathbf{w} - \varepsilon_0 \mathbf{1} \right\|_2^2 \right\}, \end{aligned} \quad (11b)$$

where $\mathcal{T}$ denotes the support set corresponding to the s largest entries in $|\mathbf{x}^{(t+1)}|$ and $\mathcal{K}$ represents the support set of the nonzero entries in $\mathbf{y}^{(t+1)}$. These subproblems are formulated by extracting the matrix columns and rows corresponding to these support sets. For example, if $\mathbf{x} = (1, 0, -3, 0, -8, 7)^\top$ and $s = 3$, then we have $\mathcal{T} = \{3, 5, 6\}$ and $\mathbf{x}_{\mathcal{T}} = (-3, -8, 7)^\top$.

The details of solving (11a) and (11b) are given as follows.

1) *Update of $\hat{\mathbf{x}}_{\mathcal{T}}$* : To obtain a tractable version of (11a), $|\mathbf{A}_{\mathcal{I}, \mathcal{T}} \mathbf{z}|$ is linearized around $\mathbf{x}_{\mathcal{T}}^{(t+1)}$. For each $i \in \mathcal{I} = \{1, \dots, m\} \setminus \mathcal{K}$, the i -th entry of $|\mathbf{A}_{\mathcal{I}, \mathcal{T}} \mathbf{z}|$ is approximated as

$$\begin{aligned} |(\mathbf{A}_{i, \mathcal{T}} \mathbf{z})| &\approx |(\mathbf{A}_{i, \mathcal{T}} \mathbf{x}_{\mathcal{T}}^{(t+1)})| \\ &\quad + \text{sgn}((\mathbf{A}_{i, \mathcal{T}} \mathbf{x}_{\mathcal{T}}^{(t+1)})) \mathbf{A}_{i, \mathcal{T}} (\mathbf{z} - \mathbf{x}_{\mathcal{T}}^{(t+1)}) \quad (12) \\ &= \text{sgn}((\mathbf{A}_{i, \mathcal{T}} \mathbf{x}_{\mathcal{T}}^{(t+1)})) \mathbf{A}_{i, \mathcal{T}} \mathbf{z}. \end{aligned}$$

This linearization preserves the local directional behavior of the magnitude of $\mathbf{A}_{\mathcal{I}, \mathcal{T}} \mathbf{z}$.

Substituting (12) into (11a) leads to a quadratic optimization problem

$$\hat{\mathbf{x}}_{\mathcal{T}}^{(t+1)} = \arg \min_{\mathbf{z} \in \mathbb{R}^{|\mathcal{T}|}} \left\{ \left\| \tilde{\mathbf{A}}_{\mathcal{I}, \mathcal{T}} \mathbf{z} - \mathbf{b}_{\mathcal{I}}^{(t+1)} \right\|_2^2 + \eta \|\mathbf{z}\|_2^2 \right\}, \quad (13)$$

where $\tilde{\mathbf{A}}_{\mathcal{I}, \mathcal{T}} = \text{diag}(\mathbf{c}_{\mathcal{I}} \odot \text{sgn}(\mathbf{A}_{\mathcal{I}, \mathcal{T}} \mathbf{x}_{\mathcal{T}}^{(t+1)})) \mathbf{A}_{\mathcal{I}, \mathcal{T}}$ and $\mathbf{b}_{\mathcal{I}}^{(t+1)} = \mathbf{c}_{\mathcal{I}} \odot \mathbf{y}_{\mathcal{I}}^{(t+1)} + \varepsilon_0 \mathbf{1}$. Then, (13) admits the closed-form solution

$$\hat{\mathbf{x}}_{\mathcal{T}}^{(t)} = (\tilde{\mathbf{A}}_{\mathcal{I}, \mathcal{T}}^\top \tilde{\mathbf{A}}_{\mathcal{I}, \mathcal{T}} + \eta \mathbf{I})^{-1} \tilde{\mathbf{A}}_{\mathcal{I}, \mathcal{T}}^\top \mathbf{b}_{\mathcal{I}}^{(t+1)}. \quad (14)$$

2) *Update of $\hat{\mathbf{y}}_{\mathcal{K}}$* : With $\hat{\mathbf{x}}_{\mathcal{T}}^{(t+1)}$, a closed-form optimal solution of (11b) is derived as

$$\hat{\mathbf{y}}_{\mathcal{K}}^{(t+1)} = \varepsilon_0 \mathbf{1} - \mathbf{c}_{\mathcal{K}} \odot |\mathbf{A}_{\mathcal{K}, \mathcal{T}} \hat{\mathbf{x}}_{\mathcal{T}}^{(t+1)}| + \mathbf{c}_{\mathcal{K}} \odot \boldsymbol{\tau}_{\mathcal{K}}. \quad (15)$$

We summarize the procedure of the proposed method in Algorithm 1. The initialization of $\mathbf{x}$ adapts the spectral method SI-1bSPR from [28].

Regarding the convergence analysis, several references have analyzed the convergence behavior of the projected gradient descent for nonconvex problems [32], [33], where they require computing the lipschitz constant using the Hessian matrix. In our problem, it is difficult to calculate the Hessian matrix due to the absolute operation. Therefore, we will consider using a surrogate function to approximate the absolute operator and then attempt to establish the convergence guarantee in our future work.

IV. NUMERICAL EXPERIMENT

In this section, we provide numerical simulation to showcase the superior performance of our algorithm over BIHT-1bSPR [28].

Algorithm 1 Projected gradient Descent and subspace pursuit for one-bit Sparse phase retrieval

Input: $\mathbf{A}$, $\mathbf{c}$, $\boldsymbol{\tau}$, s , k , β , T , J
Initialize: $\mathbf{x}^{(0)}$ by SI-1bSPR, set $\mathbf{y}^{(0)} = \mathbf{0}$.
for $t = 0 : T - 1$ **do**
 for $j = 0 : J - 1$ **do**
 Update $\mathbf{x}^{(t+1)}$, $\mathbf{y}^{(t+1)}$ via (10a), (10b) respectively;
 Set stepsize $\alpha = \frac{\beta^j}{2m}$;
 end for
 if $\mathcal{T}$ and $\mathcal{K}$ are unchanging **then**
 Update $\hat{\mathbf{x}}_{\mathcal{T}}^{(t+1)}$, $\hat{\mathbf{y}}_{\mathcal{K}}^{(t+1)}$ via (14), (15) respectively;
 Replace $\mathbf{x}_{\mathcal{T}}^{(t+1)}$, $\mathbf{y}_{\mathcal{K}}^{(t+1)}$ with $\hat{\mathbf{x}}_{\mathcal{T}}^{(t+1)}$, $\hat{\mathbf{y}}_{\mathcal{K}}^{(t+1)}$ if loss decreases.
 Break if satisfied twice consecutively.
 end if
end for
Output: $\mathbf{x}$

A. Simulation Setup

The synthetic dataset is generated as follows. The entries of $\mathbf{A} \in \mathbb{R}^{m \times n}$ follow independent and identically distributed (i.i.d.) Gaussian distribution, i.e., $a_{ij} \sim \mathcal{N}(0, 1)$. The normalized ground-truth signal $\mathbf{x}^* \in \mathbb{R}^n$ is s -sparse with nonzero entries sampled from $\mathcal{N}(0, 1)$, such that $\|\mathbf{x}^*\|_2 = 1$. The quantization threshold $\boldsymbol{\tau} \in \mathbb{R}^m$ is generated as $\tau_i = \text{median}(|\mathbf{A} \mathbf{x}^*|)$. In recovery procedure, we assume that $\boldsymbol{\tau}$ is not exactly known. Therefore, the quantization threshold is set as $(\tau_i + t \cdot z_i)$, where $z_i \sim \mathcal{N}(0, 1)$ and $t > 0$. In contrast, BIHT-1bSPR [28] adopts $\boldsymbol{\tau}$ in the recovery stage. The noise-free one-bit measurements are given by $\mathbf{c}^* = \text{sgn}(|\mathbf{A} \mathbf{x}^*| - \boldsymbol{\tau})$. To simulate measurement corruption, we introduce sign flips by randomly reversing the sign of $r \cdot m$ entries in $\mathbf{c}^*$, where r denotes the flipping ratio. Since the problem is invariant to a global phase shift, we define the aligned estimate as $\hat{\mathbf{x}} = \text{sign}(\langle \mathbf{x}, \mathbf{x}^* \rangle) \mathbf{x}$.

To evaluate the estimation performance, we use the normalized mean square error (NMSE), mean square error (MSE), and Hamming error (HE), defined as

$$\text{NMSE} = n \cdot \text{MSE} = \|\hat{\mathbf{x}} - \mathbf{x}^*\|_2^2, \quad (16)$$

$$\text{HE} = \frac{1}{m} \|\text{sign}(|\mathbf{A} \hat{\mathbf{x}}| - \boldsymbol{\tau}) - \mathbf{c}^*\|_0. \quad (17)$$

The default parameters are $(n, m, s, r) = (1000, 8n, 0.01n, 0.15)$. The proposed method uses $(\eta, \varepsilon_0, k, t) = (0.01, 1.2, 0.15m, 0.3)$. All results are averaged over 1000 Monte Carlo trials.

B. Model Parameter

This subsection analyzes the sensitivity of our algorithm to its internal parameters to determine the optimal configuration. We vary one parameter while fixing the remainder.

First, we examine η that regulates the trade-off between data fidelity and the smoothness constraint. Fig. 1 shows that reconstruction performance is stable across a wide range of η ($[10^{-4}, 1]$). Therefore, we set $\eta = 0.01$ for subsequent simulation.

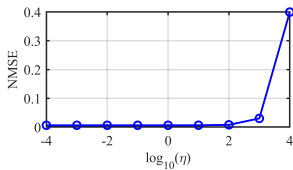


Fig. 1. Effect of η on estimation performance

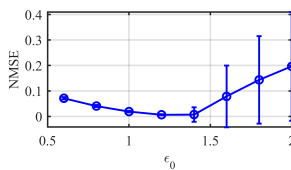
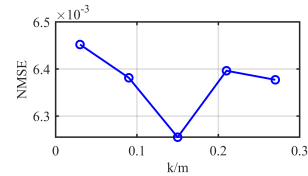
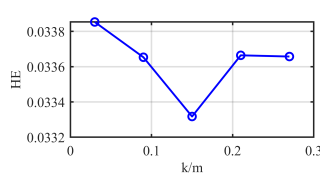


Fig. 2. Effect of ε_0 on estimation performance



(a) k ratio versus NMSE



(b) k ratio versus HE

Fig. 3. Effect of k ratio on estimation performance.

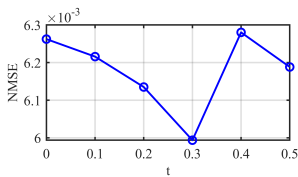


Fig. 4. Effect of t on estimation performance

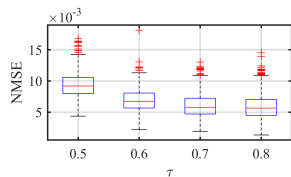


Fig. 5. Effect of τ on estimation performance

Next, we assess the effect of ε_0 on the estimation performance. As shown in Fig. 2, NMSE achieves optimal performance with around $\varepsilon_0 = 1.2$. Consequently, we set $\varepsilon_0 = 1.2$.

Then, the impact of k is investigated. As depicted in Fig. 3, both NMSE and HE remain nearly constant over the tested range. This observation suggests that GSPR maintains stable performance even under sparsity mismatch conditions. We set $k = 0.15m$ for the following simulation.

Prior studies [34], [35] have utilized random dithers to mitigate quantization loss and to improve the estimation of signal characteristics during coarse quantization. Then, we investigate the effect of threshold equipped with random dithers. As depicted in Fig. 4, the NMSE of our method remain constant as t increases. This indicates that the proposed method exhibits empirical robustness to mild threshold dithers. Therefore, we fix $t = 0.3$ in all subsequent experiments.

Moreover, to evaluate the performance without relying on $\mathbf{x}^*$, we use the fixed constant thresholds ($\tau \in \{0.5, 0.6, 0.7, 0.8\} \cdot 1$). As shown in Fig. 5, the reconstruction errors remain low across these fixed values. These results indicate that fixed τ is feasible, clarifying that the selection of τ does not depend on $\mathbf{x}^*$.

C. Numerical Comparison

First, we compare the reconstruction performance under different sampling ratios m/n . As shown in Fig. 6, both methods achieve lower MSE with increasing sampling ratio.

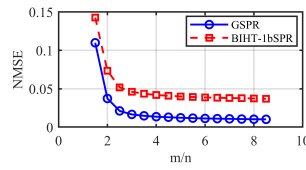


Fig. 6. NMSE versus m/n

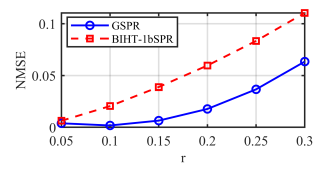


Fig. 7. NMSE versus r

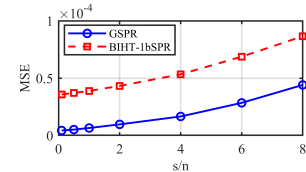


Fig. 8. MSE versus s/n

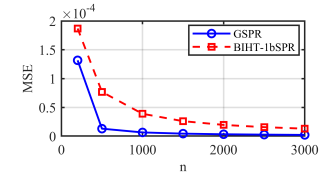


Fig. 9. MSE versus n

The proposed method attains higher accuracy than BIHT-1bSPR.

Then, the performance of both approaches under different r is investigated in Fig. 7. Although their performance degrades as r increases, our approach maintains lower NMSE. This confirms that our method effectively mitigates the impact of noise.

Subsequently, we compare them under different signal sparsity. Fig. 8 shows that their MSEs increase with s/n raising. However, the suggested algorithm maintains a lower reconstruction error than BIHT-1bSPR across all sparsity levels. Even as the sparsity level of the signal decreases, our method demonstrates a distinct performance advantage, confirming its robustness to varying sparsity conditions.

Furthermore, we compare the recovery performance of different methods with respect to n . As shown in Fig. 9, GSPR consistently outperforms BIHT-1bSPR as n increases, verifying its effectiveness in high-dimensional settings.

V. CONCLUSION

In this work, the one-bit sparse phase retrieval problem was formulated as a double-sparsity constrained optimization task. By introducing a sparse auxiliary variable, we explicitly modeled sign-inconsistency to mitigate the impact of noise. Then, we devised an algorithm that integrated gradient projection with backtracking and subspace pursuit to tackle the resultant nonconvex optimization problem. Specifically, we employed a first-order linearization to approximate the absolute-value term, converting the subproblem into a tractable least-squares formulation with a closed-form solution. Numerical results verified that the proposed method yielded superior reconstruction accuracy over the state-of-the-art method in the presence of noise.

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